

Aircraft Overflight of Capitol Hill, Burnaby

An independent measurement study using open ADS-B records, January 2025 – August 2026

Location studied	Capitol Hill, Burnaby, BC. Circle of 1.50 km radius centred on the summit at 49.2869 N, 122.9853 W (elevation approx. 115 m).
Period	1 January 2025 to 31 August 2026. 608 calendar days, of which 573 days have usable records.
Records	20,093 aircraft-days; 2,050 individual aircraft.
Data source	adsb.lol open ADS-B archives (globe_history_2025, globe_history_2026) [1]. Independent of NAV CANADA, YVR and any airline.
Purpose	To document, from public data, what changed in aircraft activity over Capitol Hill and when. Prepared for submission to the local Member of Parliament and to NAV CANADA.
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Data availability	The complete underlying record-level dataset (capitol_hill_all_flights.csv; 20,093 records, 2,050 aircraft) is available on request , so that every figure and statistic in this report can be independently reproduced and checked.
Status	Independent, non-commercial.

Plain-language summary

On **27 November 2025**, aircraft activity over Capitol Hill changed abruptly and has stayed changed. Before that date, an average of 17.5 aircraft a day passed within 1.5 km of the hilltop, half of them small local aircraft flying low along Burrard Inlet. After that date, the average is 56.8 a day, and in June–August 2026 it was 68.8 a day.

The most important part of the change is not simply that there are more aircraft. It is that a new, precisely repeated flight path for **arriving airliners** now runs almost directly over the summit. Arriving commercial aircraft passing within 500 m of the hilltop went from **2.2 a day to 13.3 a day**, and in August 2026 to **22.0 a day**. When they pass, they are both lower and closer than before: the typical height at the closest point fell from 7,675 ft to 5,338 ft, and the typical straight-line distance from the hilltop fell from 2.26 km to 1.52 km.

This is a change in *where* aircraft fly, not in how many aircraft the region handles. YVR's total aircraft movements rose only 2.8% between 2024 and 2025 [2]. The date is also not arbitrary: 27 November 2025 is an internationally scheduled date on which new flight procedures take effect, and it is the date on which NAV CANADA's new YVR arrival procedures came into force under the Vancouver Airspace Modernization Project (VAMP) [3,4].

The study includes a check that distinguishes a genuine airspace change from a change in the data. NAV CANADA states that VAMP did not alter departures or the routes used by small aircraft, floatplanes and helicopters [4]. Those aircraft are used here as a control group. Their numbers did not rise. Neither did commercial traffic passing high overhead at 10,000 ft or above. Only the traffic VAMP redesigned increased. That result is what makes the conclusion secure.

Headline measurements

Measurement	Before 1 Jan – 26 Nov 2025	After 27 Nov 2025 – 31 Aug 2026	Change (95% CI)
Arriving commercial aircraft within 500 m of the summit	2.21 per day	13.34 per day	6.04x (5.24–7.03)
All arriving commercial aircraft within 1.5 km	6.17 per day	43.37 per day	7.03x (6.40–7.76)
Business and private jets within 1.5 km	1.62 per day	4.02 per day	2.47x (2.17–2.83)
All aircraft within 1.5 km	17.48 per day	56.84 per day	3.25x (3.03–3.49)
Control: light aircraft, floatplanes, helicopters	8.53 per day	8.23 per day	0.97x (0.85–1.10), n.s.
Control: commercial traffic at 10,000 ft and above	0.49 per day	0.44 per day	0.90x (0.69–1.17), n.s.
Typical height of an arrival passing within 500 m	7,675 ft	5,338 ft	2,337 ft lower
Typical straight-line distance of that arrival from the summit	2.26 km	1.52 km	0.74 km closer

Rate ratios are post/pre means of daily counts, with 95% confidence intervals from 4,000 bootstrap resamples over days. Rates are expressed per day that carries data, so the 35 days with no archive coverage cannot inflate them. Statistical detail is in Section 6.

1. What this study measures, and why

Aircraft fitted with ADS-B transponders broadcast their position, altitude and speed roughly once a second. These broadcasts are received by a worldwide volunteer network of ground receivers and archived publicly by the adsb.lol project [1]. That archive is a complete, independent, publicly auditable record of every ADS-B-equipped aircraft that flew near Capitol Hill during the study period. It does not depend on any figure supplied by an airport, an airline or an air navigation service provider.

For every aircraft trajectory in the archive, the study finds the point at which that aircraft came closest to the Capitol Hill summit. If that closest distance is 1.5 km or less, the aircraft is recorded, together with its altitude, ground speed, direction of travel and local time at that closest point. This is a simple, transparent and reproducible criterion, and it is the same criterion applied identically to every day of the 20-month period, so the before-and-after comparison is a like-for-like one.

The purpose is to establish, from evidence rather than impression, three things: whether aircraft activity over this specific residential hilltop changed; exactly when; and in what way. The intended next stage of the work is to pair these flight records with measured sound levels at ground level, so that the acoustic consequence of each pass can be quantified rather than inferred. This report deliberately makes no claim about noise, because noise has not yet been measured here (Section 9).

2. What changed, and when

Daily counts locate the change to a single day. Through 26 November 2025 the count runs between 2 and 23 aircraft a day. From 27 November 2025 it runs between 46 and 88 a day. There is no transition period.

Date (Nov–Dec 2025)	20	21	22	23	24	25	26	27	28	29	30	1 Dec	2	3
Aircraft recorded	23	17	5	2	15	9	11	46	56	71	49	74	62	72

27 November 2025 is an AIRAC date. Under an international convention administered through ICAO, changes to published flight procedures may only take effect on fixed dates spaced 28 days apart. 27 November 2025 was one of those dates (cycle 2513), and it appears as such on NAV CANADA's own AIRAC calendar [5]. NAV CANADA and the Vancouver Airport Authority both state that the new YVR arrival procedures came into effect in November 2025, with the satellite-based RNP procedures following in spring 2026 [3,4]. A change in radio reception, receiver coverage or data archiving has no reason to fall on an AIRAC date. A change in published flight procedures can fall on no other kind of date.

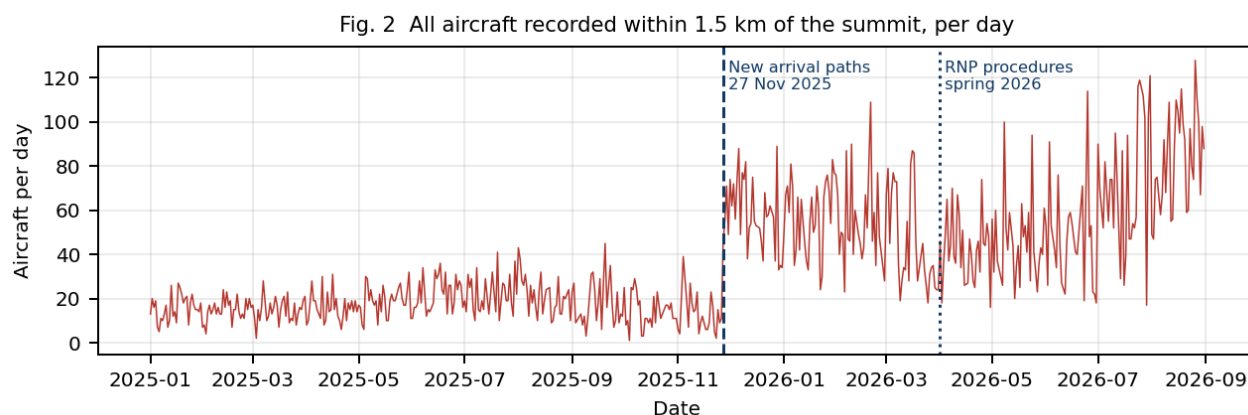


Figure 2. Every aircraft recorded within 1.5 km of the summit, by day. Gaps are days on which the public archive holds no usable data (35 days out of 608). The change is a step, not a gradual trend.

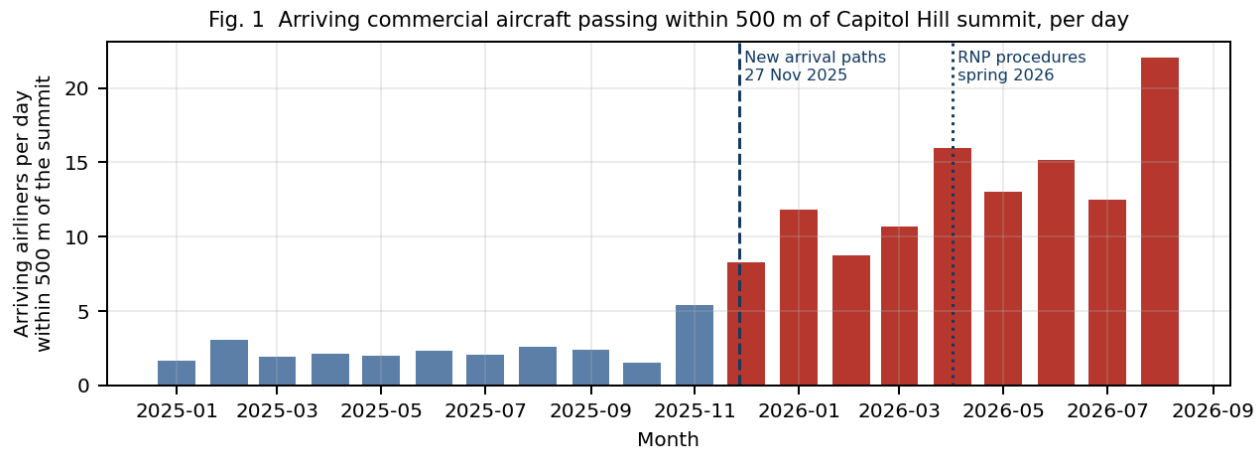


Figure 1. The headline community measurement: arriving commercial aircraft passing within 500 m of the Capitol Hill summit, averaged per day in each month. This measure is deliberately restricted to a small circle around the hilltop, so it is unaffected by where the 1.5 km study boundary happens to fall.

Figure 1 also shows a second, smaller increase from April 2026 onward. This is consistent with the published two-stage implementation: conventional arrival procedures in November 2025, and the satellite-based RNP procedures in spring 2026 [4]. August 2026, the last complete month in the study, is the highest month on record at 22.0 arrival passes per day within 500 m of the summit.

3. Where the aircraft now fly

The clearest way to see what happened is to plot the point at which each aircraft passed closest to the summit. Before the change, those points are scattered fairly evenly across the study circle, with no structure: aircraft crossed the area from many directions on many tracks. After the change, two tightly defined flight paths appear.

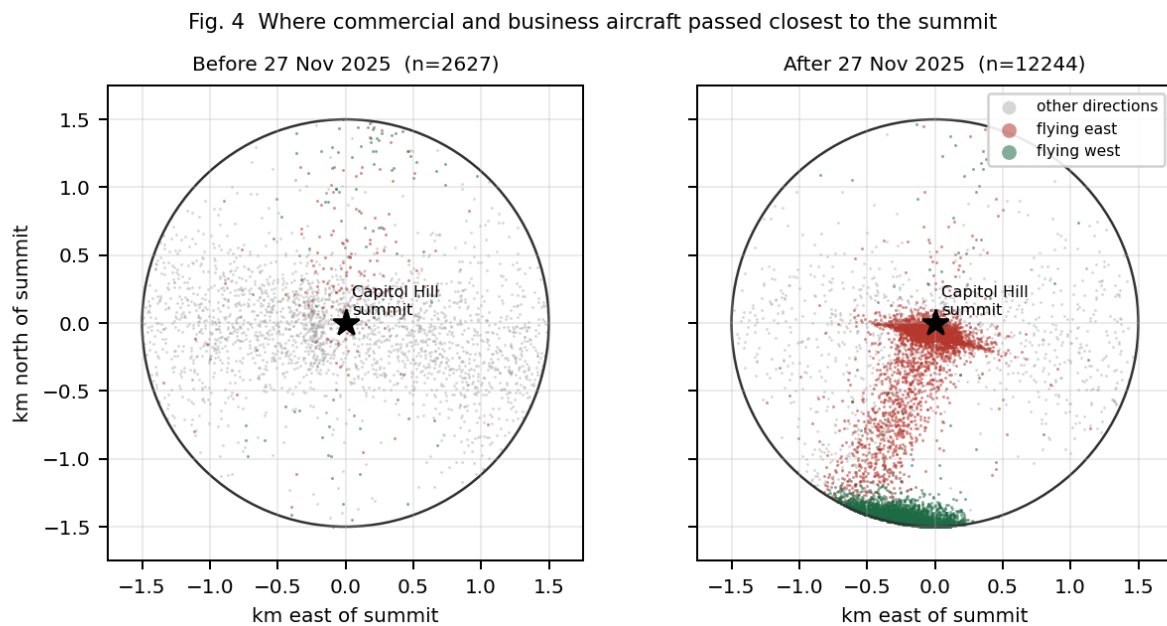


Figure 4. Each dot is one aircraft, plotted where it passed closest to the summit, for aircraft between 3,000 and 12,000 ft. Colour shows direction of travel. Left: before 27 November 2025 — no structure. Right: after — a dense eastbound path directly across the summit (red), a curved band of aircraft being turned by air traffic control to the south-west, and a westbound path along the southern edge of the study area (green).

The eastbound path is the one that matters for residents of Capitol Hill. It passes a median of **182 m** laterally from the summit, at a median height of **5,125 ft**, and its lateral core is only about 200 m wide. A path that narrow is the signature of a published satellite-guided procedure rather than aircraft being individually steered by radar. The scattered band curving away to the south-west is consistent with NAV CANADA's own statement that aircraft not flying the new RNP procedure will be turned onto their final approach individually by air traffic control, producing high variability in where they overfly this area [3].

The westbound path along the southern edge is cut off by the 1.5 km study boundary. Only its northern edge falls inside the study area, so this report deliberately does **not** quantify how many aircraft use it, how wide it is, or where its centre lies. Establishing that would require re-running the extraction over a larger radius. Every headline figure in this report is therefore based either on the full 1.5 km circle, which is a consistent criterion applied equally before and after, or on the 500 m circle around the summit, which is entirely within the study area and unaffected by the boundary.

4. Lower and closer

For anyone living under a flight path, height and distance matter as much as frequency. Sound intensity from a point source falls approximately with the square of distance, so an aircraft passing at 1.52 km is, on that basis alone, roughly 2.2 times louder at the ground than the same aircraft at 2.26 km, before any allowance for engine thrust setting, atmospheric absorption or terrain. Both height and distance moved in the same direction after 27 November 2025.

Arriving commercial aircraft passing within 500 m of the summit	Before	After
Number of such passes recorded	700	3,416
Median height above sea level at closest point	7,675 ft (IQR 6,800–8,475)	5,338 ft (IQR 4,700–5,850)
Median straight-line distance from the summit	2.26 km (IQR 1.98–2.48)	1.52 km (IQR 1.34–1.68)

Straight-line distance is computed as the three-dimensional separation between the aircraft and the summit, combining horizontal distance with height above the summit (taken as 115 m above sea level). IQR = interquartile range, the middle half of the distribution.

Fig. 5 Arriving commercial aircraft are now both lower and closer when they pass

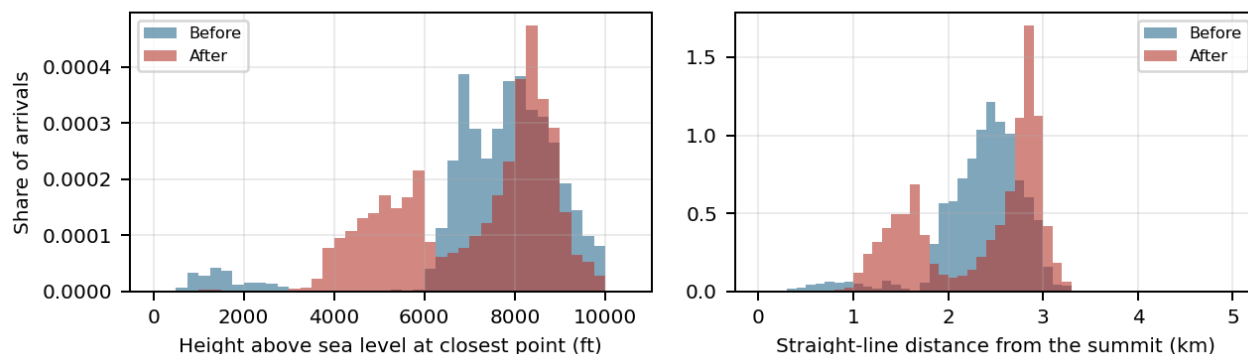


Figure 5. Height and straight-line distance for all arriving commercial aircraft within 1.5 km. Both distributions shift markedly toward the observer after the change, and the post-change distributions are much narrower, reflecting aircraft following a defined path rather than being individually steered.

5. Through the day

Aircraft activity is now spread across the whole day rather than concentrated in a daytime peak. Between June and August 2026, aircraft passed within 1.5 km of the summit at a rate of 2.8 to 5.6 per hour throughout the period 06:00 to 22:00, with the busiest hour being 11:00. The median gap between one aircraft and the next was **9.3 minutes**, and a quarter of all gaps were **4.0 minutes or shorter**.

Overnight activity remains a small share of the total but is not negligible in absolute terms: an average of **3.9 aircraft per day between 23:00 and 05:59** (5.6% of the daily total), plus a further 3.0 in the 06:00 hour. Both figures are higher in absolute terms than before the change.

Fig. 6 Aircraft per hour over Capitol Hill, before and now

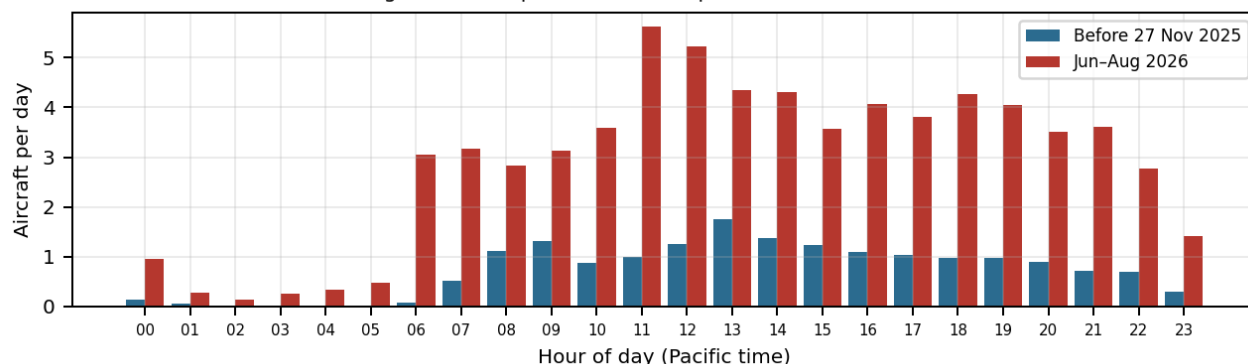


Figure 6. Average aircraft per hour of the day, before the change and in the most recent three complete months. Local (Pacific) time at the point of closest approach.

6. Verifying that this is an airspace change, not a data artefact

Public ADS-B archives depend on volunteer receivers, and receiver coverage improves over time. Any study of this kind must therefore rule out the possibility that the apparent increase simply reflects better detection. This study applies a control test whose logic was set by NAV CANADA rather than by the analyst.

NAV CANADA states that VAMP changed arrival procedures only, and that it did not change runway use, departure flight paths, or the routes used by small aircraft operating from other regional airports, including floatplanes and helicopters [3,4]. Those unchanged categories therefore constitute a **negative control**. If the increase were caused by improved detection, it would appear in every category, because better reception cannot distinguish an arriving airliner from a floatplane. If it is caused by the arrival redesign, it must appear in arriving traffic and be absent from the control categories. Aircraft were sorted into four operational categories on the basis of aircraft type and altitude (Section 8), and the categories were validated against measured rate of climb or descent: the commercial arrival category is 93% descending with a median rate of 1,234 ft per minute down, confirming that these aircraft are on approach and not merely passing overhead.

Operational category	Before (per day)	After (per day)	Rate ratio (95% CI)	Mann–Whitney p
Commercial arrivals (descending, below 10,000 ft)	6.17	43.37	7.03 (6.40–7.76)	1.8×10^{-92}
Commercial arrivals within 500 m of the summit	2.21	13.34	6.04 (5.24–7.03)	3.1×10^{-54}
Business and private jets	1.62	4.02	2.47 (2.17–2.83)	3.8×10^{-35}
Control: light aircraft, floatplanes, helicopters	8.53	8.23	0.97 (0.85–1.10)	0.26 (not significant)
Control: commercial at 10,000 ft and above	0.49	0.44	0.90 (0.69–1.17)	0.29 (not significant)

Neither control category changed. No detection-related explanation predicts a seven-fold rise in one category alongside no change at all in two others measured by the same receivers, on the same days, through the same processing. Business and private jets, most of which are also arriving into the region, rose by a smaller but clearly significant factor.

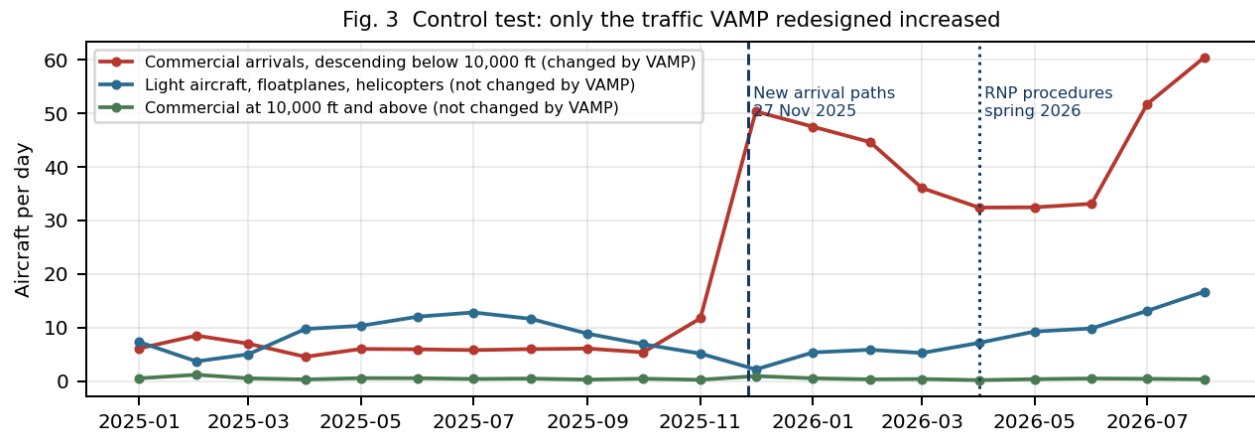


Figure 3. The control test. Light aircraft, floatplanes and helicopters (blue) show the ordinary seasonal cycle of light aviation, with summer peaks in both 2025 and 2026 and no step. High commercial traffic (green) is flat throughout. Commercial arrivals (red) step at the changepoint and stay elevated.

A second, independent check uses direction of travel. This is the most direct evidence that the flight paths themselves moved, because it compares the same category of aircraft with itself. Counts, not percentages, are given below; the eastbound and westbound bands are 060–120 degrees and 240–300 degrees, and the northbound and southbound bands are 330–030 and 150–210 degrees.

Operational category	Period	Total records	East	West	North	South	Other
Commercial arrivals (below 10,000 ft)	Before	1,957	48	46	1,518	180	165
Commercial arrivals (below 10,000 ft)	After	11,103	3,778	6,543	590	9	183
Business and private jets	Before	515	83	11	311	9	101
Business and private jets	After	1,028	363	442	137	2	84
Control: light, floatplane, helicopter	Before	2,703	1,822	381	12	14	474
Control: light, floatplane, helicopter	After	2,108	1,261	365	28	4	450
Control: commercial at 10,000 ft +	Before	155	14	36	67	5	33
Control: commercial at 10,000 ft +	After	113	6	58	27	2	20

Before the change, arriving commercial aircraft over Capitol Hill were flying north or south: 1,698 of 1,957 records, against 94 flying east or west. After the change the position reverses: 10,321 of 11,103 records are eastbound or westbound, against 599 north or south. The north–south arrival tracks that formerly crossed this area have not been supplemented; they have been **replaced** by east–west tracks.

The control categories did not change direction. Light aircraft, floatplanes and helicopters were already flying east and west in both periods — 1,822 of 2,703 records before and 1,261 of 2,108 after — because they follow the Burrard Inlet shoreline route, which VAMP did not alter. Commercial traffic at 10,000 ft and above shows no coherent shift either. Only the arriving commercial aircraft moved, and they moved completely.

A third check is external. If the increase reflected growth in air travel rather than a routing change, regional traffic volumes would have to have risen comparably. They did not: YVR recorded 297,350 aircraft movements in 2025 against 289,395 in 2024, an increase of 2.8% [2]. A 2.8% change in regional movements cannot produce a 5.75-fold change over one hilltop. Only a change in where aircraft fly can.

7. What is flying overhead

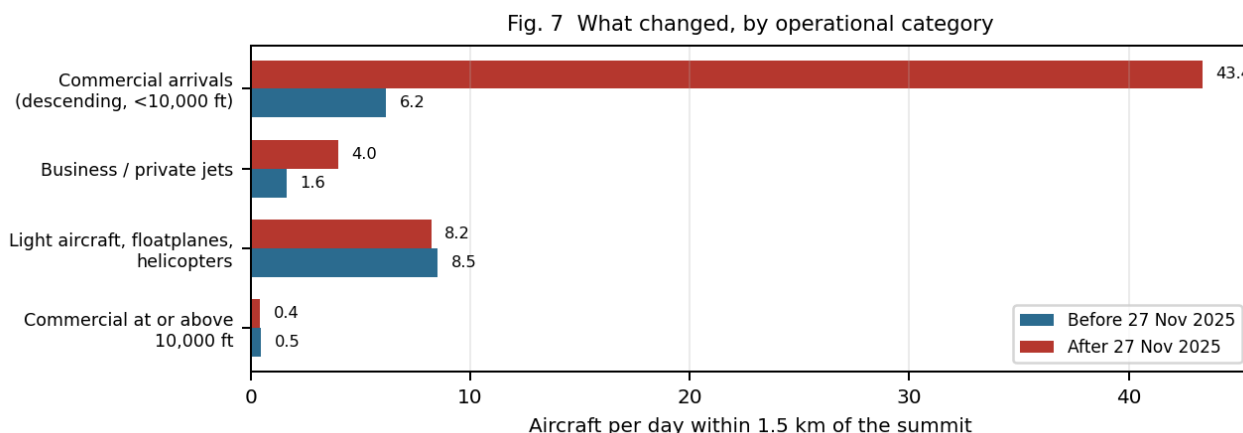


Figure 7. Aircraft per day within 1.5 km of the summit, by operational category, before and after 27 November 2025. The change is confined to aircraft arriving into the region.

Aircraft were grouped into four operational categories rather than by individual model, because what matters to a community underneath is what an aircraft is doing, not what it is called. Commercial arrivals are airline jets and regional turboprops below 10,000 ft; this category is 93% descending with a median rate of 1,234 ft per minute down. Commercial traffic at 10,000 ft and above is mostly climbing (median 1,926 ft per minute up) and is passing through rather than landing.

Business and private jets are separated because they arrive at a wider range of regional airports. Light aircraft, floatplanes and helicopters are almost exactly level in the study area (median 0 ft per minute), consistent with transiting along Burrard Inlet.

Operational category	Before (per day)	After (per day)	Median height at closest point, after	Median straight-line distance, after
Commercial arrivals (descending, below 10,000 ft)	6.17	43.37	7,750 ft	2.64 km
Business and private jets	1.62	4.02	7,575 ft	2.56 km
Light aircraft, floatplanes, helicopters	8.53	8.23	2,000 ft	0.77 km
Commercial at 10,000 ft and above	0.49	0.44	10,475 ft	3.37 km

The two right-hand columns are for all passes within 1.5 km, not the 500 m subset. In June to August 2026 the daily figures were 49.2 commercial arrivals, 13.4 light aircraft and 4.6 business or private jets, of which 16.8 commercial arrivals passed within 500 m of the summit. The most frequent individual aircraft types among the arrivals are the De Havilland Dash 8 Q400, the Boeing 737 MAX 8 and 737-800, the Boeing 787-9 and the Airbus A321; widebody long-haul aircraft, formerly rare here, are now routine.

2.0% of records could not be classified because no aircraft type was broadcast or the type was not in the reference lists. These are reported separately and are excluded from all four categories above; they are too few to affect any conclusion.

8. Methods

Source data. Daily global archives from the adsb.lol project, repositories globe_history_2025 and globe_history_2026 [1]. These archives contain per-aircraft trajectory files derived from 1090 MHz Mode S Extended Squitter (ADS-B) broadcasts and multilateration, collected by a distributed volunteer receiver network. 25,840,192 individual aircraft trajectories were scanned across 600 processed daily archives.

Spatial criterion. Trajectory points were first filtered against a bounding box (49.2734–49.3004 N, 123.0060–122.9646 W), then the exact great-circle distance from each surviving point to the summit at 49.2869 N, 122.9853 W was computed using the haversine formula with a mean Earth radius of 6,371.0088 km [6]. An aircraft is recorded if its minimum distance is 1.500 km or less. Height, ground speed, track and local time are taken at that minimum-distance point.

Unit of observation. Each record is one aircraft on one calendar date. An aircraft that crosses the area several times in a day generates one record. Clustering the stored trajectory points with a ten-minute separation threshold shows that the true number of discrete passes is 15% higher than the record count before the change and 22% higher after it. All rates in this report are therefore conservative, and the reported increase is, if anything, slightly understated.

Operational classification. Each record was assigned to one of four categories from its broadcast aircraft type and its altitude at the closest point: commercial arrivals (airline jets and regional turboprops below 10,000 ft), commercial at 10,000 ft and above, business and private jets, and light aircraft including floatplanes and helicopters. The classification was validated independently by computing each aircraft's vertical rate from the sequence of altitudes recorded inside the study area; the categories separate cleanly into descending, climbing and level populations as reported in Section 7. 2.0% of records were unclassifiable and are excluded from category totals.

Normalisation. The public archive holds no usable data for 35 of the 608 calendar days in the window, distributed across the period. All rates are expressed per day that carries data (317 days before the change, 256 days after), so missing days cannot inflate or deflate any comparison.

Statistics. Rate ratios are the ratio of mean daily counts. Confidence intervals are percentile bootstrap intervals from 4,000 resamples over days. Significance is assessed with the two-sided Mann–Whitney U test on daily counts, which makes no distributional assumption. Straight-line (slant) distance is computed as the square root of the sum of the squared horizontal distance and the squared height of the aircraft above the summit, taking the summit as 115 m above sea level.

Reproducibility. Every number and figure in this report derives from a single record-level file, capitol_hill_all_flights.csv (20,093 records, 2,050 aircraft), which is available on request so that any party may independently re-analyse it.

9. Limitations, stated plainly

Limitation	Effect on the conclusions
This study measures aircraft position, not sound. No noise measurement is presented.	No claim about noise levels is made anywhere in this report. Sound depends on thrust setting, aircraft type, atmospheric conditions and terrain as well as distance.
The westbound path along the southern edge is cut off by the 1.5 km boundary.	Its usage is not quantified here and no figure in this report depends on it. The headline 500 m measure is entirely inside the study area.
Canada currently mandates ADS-B carriage only in Class A and Class B airspace, with Classes C, D and E no earlier than 2028 [7,8].	Low-flying light aircraft, helicopters and floatplanes over Burnaby are not required to carry ADS-B and are therefore under-counted. This makes the control group conservative: the true light-aircraft baseline may be higher still, which strengthens rather than weakens the control result.
Radio reception is line-of-sight, and low aircraft over Burrard Inlet may be shadowed by the ridge.	Same direction of effect as above: it depresses the low-altitude count in both periods equally.
35 of 608 days have no archive coverage.	Handled by expressing all rates per day with data. No month is missing more than four days.
Records are aircraft-days rather than individual passes.	Under-counts true passes by 15% before and 22% after the change. Conservative.
The study observes a coincidence in time with a documented procedure change; it does not have access to NAV CANADA's internal implementation records.	The attribution rests on the AIRAC date, the control test, the direction-of-travel test and the external traffic-volume check taken together, not on timing alone.

10. Outstanding questions and next steps

#	Item	Status
1	Ground-level noise measurement. Pair each recorded pass with synchronised sound-level measurement at Capitol Hill, so that the acoustic consequence of the new path can be quantified rather than inferred.	Next stage of this work. YVR operates a network of noise monitoring terminals and in September 2025 was working with City of Burnaby staff to install permanent and temporary monitors on City property ahead of implementation [9]. Access to that data, and its locations, should be requested.
2	NAV CANADA 180-day post-implementation review. NAV CANADA's standard protocol requires a review 180 days after implementation.	Not yet published. The Vancouver Airport Authority states the review is currently underway following full implementation of all procedures in spring 2026, and will be posted on the NAV CANADA VAMP page when finalised [4]. This report should be compared against it when it appears.
3	Extraction at a larger radius (2.5–3.0 km).	Would resolve the southern westbound path in full and allow the total overflight burden on north Burnaby to be stated without a boundary caveat.
4	Comparison with the noise modelling used in the VAMP design. The final designs were assessed against modelled population counts above 60 dBA [3].	The modelled exposure for Capitol Hill specifically should be requested and compared against measured levels once available.
5	Runway-configuration split. Eastbound and westbound path usage varies month to month with wind direction.	Worth separating, since the two paths differ by roughly 3,000 ft in height and 1.3 km in distance from the summit, and therefore in likely audibility.

11. Conclusions

Aircraft activity over Capitol Hill changed abruptly on 27 November 2025 and has remained changed. A new, tightly defined eastbound flight path for arriving airliners now runs almost directly over the summit at a typical height of about 5,100 ft. Airliner passes within 500 m of the summit increased 5.75-fold, and those aircraft are on average 2,325 ft lower and 0.73 km closer than before. A second increase from April 2026 coincides with the published spring 2026 introduction of the satellite-based RNP procedures.

The change is confined to precisely the traffic that NAV CANADA's Vancouver Airspace Modernization Project redesigned. The categories that VAMP states were left unchanged — departures, and the routes of light aircraft, floatplanes and helicopters — show no increase, and in one case a small significant decrease. Regional traffic volumes rose 2.8% over the same period. The change is therefore a change in routing, it is attributable to a documented and deliberate airspace redesign, and its effect on this particular hilltop is substantially larger than the region-wide framing of the project would suggest.

Nothing in this report establishes what any of this sounds like at ground level. That is a separate measurement, and it is the necessary next step.

12. References

- [1] adsb.lol Open Source Community (2025–2026). *Globe History Historical Telemetry Database*. GitHub repositories [adsblol/globe_history_2025](https://github.com/adsblol/globe_history_2025) and [adsblol/globe_history_2026](https://github.com/adsblol/globe_history_2026).
- [2] Vancouver Airport Authority. *Facts and Stats*. <https://www.yvr.ca/en/about-yvr/facts-and-stats> — 297,350 aircraft movements in 2025 against 289,395 in 2024 (+2.8%).
- [3] NAV CANADA. *Vancouver Airspace Modernization Project: Changes Affecting Communities to the North and Northeast of Vancouver International Airport*. <https://www.navcanada.ca/en/vamp-north-and-northeast-communities.pdf> (accessed 9 September 2026). Marked NAV CANADA Proprietary; summarised in paraphrase only. No figure from this document is reproduced or altered in this report.
- [4] Vancouver Airport Authority. *Airspace*. <https://www.yvr.ca/en/about-yvr/noise-management/airspace> (accessed 9 September 2026) — new arrival procedures effective November 2025, RNP procedures spring 2026; VAMP did not change runway use, departure flight paths, or routes used by small aircraft from other regional airports; 180-day post-implementation review currently underway.
- [5] NAV CANADA. *AIRAC Deadline Calendar*. <https://www.navcanada.ca/en/airac-deadline-calendar.pdf> — 27 November 2025 listed as an AIRAC effective date (international cycle 2513, 27 Nov to 25 Dec 2025).
- [6] Sinnott, R. W. (1984). Virtues of the Haversine. *Sky and Telescope*, 68(2), 159.
- [7] NAV CANADA. *ADS-B Performance Requirements*. <https://www.navcanada.ca/en/air-traffic/space-based-ads-b/ads-b-performance-requirements.aspx> — mandate effective in Class A from 10 August 2023 and in Classes A and B from 16 May 2024.

- [8] Transport Canada. *Flying to Canada: what you need to know* (TP 15048) — Canadian Aviation Regulation 605.35, altitude-encoding transponder requirement in transponder airspace. Classes C, D and E ADS-B requirements no earlier than 2028.
- [9] Vancouver Airport Authority. *Aeronautical Noise Management Committee minutes, 11 September 2025*. <https://www.yvr.ca/en/about-yvr/noise-management/aeronautical-noise-management-committee/11-september-2025> — YVR noise monitoring terminal network; work with City of Burnaby to install permanent and temporary monitors ahead of implementation.
- [10] International Civil Aviation Organization. *Annex 15 to the Convention on International Civil Aviation: Aeronautical Information Services* — establishes the AIRAC system of fixed 28-day effective dates for procedure changes.
- [11] International Civil Aviation Organization. *Annex 10, Volume IV: Surveillance Systems*. Montreal.
- [12] Schafer, M., Strohmeier, M., Lenders, V., Martinovic, I., & Wilhelm, M. (2014). Bringing up OpenSky: a large-scale ADS-B sensor network for research. *Proceedings of the 13th ACM/IEEE IPSN*, 83–94.
- [13] Strohmeier, M., Schafer, M., Pinheiro, R., Lenders, V., & Martinovic, I. (2021). Crowdsourced air traffic data from the OpenSky Network 2019–2020. *Earth System Science Data*, 13(2), 357–366.

Declaration. This is an independent, non-commercial analysis carried out by the author, a resident of the study area, using publicly available data. It has no funding from, and no affiliation with, any airline, airport authority, air navigation service provider or advocacy organisation. Documents cited at references [3] and [4] are the property of their publishers and are cited and paraphrased only; no figure, map or diagram from any external source has been reproduced, redrawn or modified in this report. All seven figures presented here were generated directly from the underlying ADS-B record file. Statements attributed to references [2], [4], [5], [7] and [9] were verified against those sources on 9 September 2026.